

EGR314 — FALL 2026 · TEAM TRENCHLESS

Sentinel: Ideation & Concept Generation

Underground Pipeline Inspection: Embedded Sensing and Control for Miniature Inspection Robots

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a. TEAM ORGANIZATION

Who we are & how we work

Jon	Internet-based two-way comms
Kennna	Human-machine interface
Ayush	Sensor & actuator control
Ty	Power & mechanical

Charter

Weekly syncs, shared GitHub repo, equal ownership of one subsystem each, and consensus before any change that affects another teammate's work.

Mission Statement

Give pipe inspectors reliable, real-time visibility into underground pipe conditions — improving defect detection while reducing manual inspection time and risk.

b. PROJECT SUMMARY

Underground Pipeline Inspection

What we learned from the talk / site visit

- Operators most often flag cracks, joint offsets, root intrusion, and corrosion — visual sensing is the highest-value capability
- Runs are staged from a surface access point with the robot pulled/pushed through on a tether, so a wired link is standard practice, not a limitation
- Pipes are pitch dark and often damp, so onboard lighting and moisture-sealed electronics are non-negotiable design constraints

Goal of our device

An embedded sensing & control system for a miniature robot that navigates constrained pipes, collects inspection data, and helps identify and localize defects.

Audience

Inspection technicians & operators (primary); municipalities, utility owners, and contractors (secondary, defect reports); research teams (sensing methods).

How we organized our brainstorm

The team met on a shared online whiteboard and generated 22 ideas against three prompts, following standard brainstorming rules: don't criticize, encourage variety, build on ideas, and don't fear the unrealistic.

Sensing & defect detection

Camera + LED ring, IMU tilt sensing, ultrasonic distance, motor-current stall detection, moisture sensor

Mobility & chassis

Tracked treads vs. wheels, adjustable spring-loaded diameter, differential drive for junctions

Comms & HMI

Tethered wired feed, onboard SD logging, status LEDs, defect-flag button, quick-start card

Power & safety

Swappable battery, auto low-battery return, IP-rated sealed housing, compliant bumper

c.ii DESIGN IDEATION

Initial capture — mid-brainstorm

20 ideas captured on our shared board, before any sorting or ranking.

IMU tilt sensing

Tracked treads

Tethered wired
feed

Swappable
battery pack

Camera +
LED ring

Peristaltic
locomotion

Status LED ring

IP-rated sealed
housing

Ultrasonic
distance sensor

Differential-drive
turning

Onboard SD
logging

Compliant
bumper/tread

Moisture/leak
sensor

Wheeled option

Defect-flag
button

Auto low-battery
return

Motor-current
stall sensing

Adjustable-
diameter chassis

Quick-start card

Tether-delivered
power

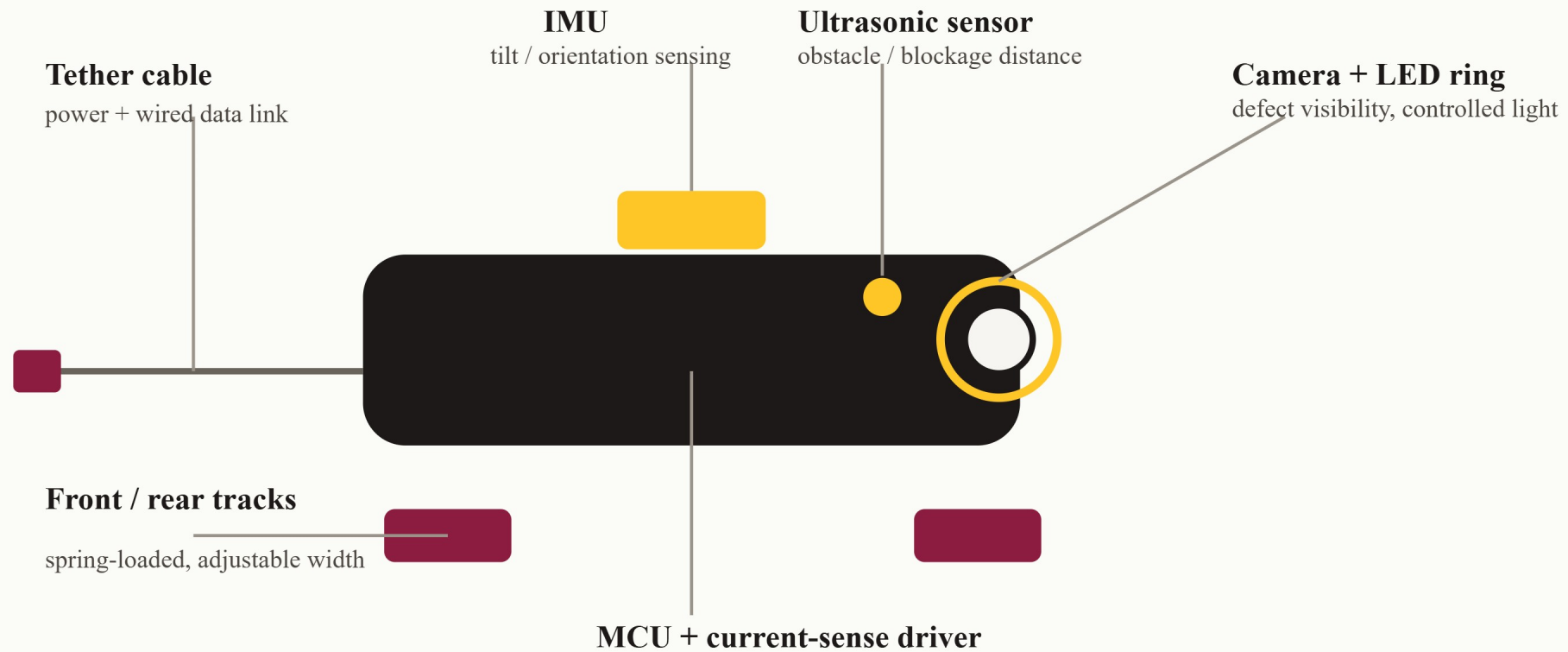
c.iii DESIGN IDEATION

Final sorted, ranked & grouped board

Sensing & Detection	Mobility & Chassis	Comms & HMI	Power & Safety
★ Camera + LED ring	★ Adjustable-diameter chassis	★ Tethered wired feed	★ IP-rated sealed housing
★ Motor-current stall sensing	Tracked treads	★ Defect-flag button	Tether-delivered power
IMU tilt sensing	Wheeled option	Status LED ring	Swappable battery pack
Ultrasonic distance sensor	Differential-drive turning	Onboard SD logging	Auto low-battery return
Moisture/leak sensor	Peristaltic locomotion	Quick-start card	Compliant bumper/tread

Gold ★ = top-ranked, recombined into the Sentinel concept

Sentinel — annotated concept



d.ii RATIONALE

Why Sentinel meets the need

Cues for easier use

A color-coded status LED ring (moving / stuck / defect found / low battery) and an audible chirp on defect detection remove the need to watch a screen constantly.

Controls design

A handheld controller with joystick drive, one defect-flag button, and an OLED status readout — few controls, single-purpose, hard to mis-press in the field.

Durability, safety, comfort

IP-rated sealed housing for moisture, a compliant bumper to avoid scraping pipe walls, and current-limit protection on the motor driver.

Instruction needed

A laminated quick-start card with the LED legend — no manual required for a technician to deploy it in the field.

The tethered wired link was chosen over wireless because underground pipes have limited connectivity. Core hardware: ESP32-WROOM-32 MCU, OV2640 camera with a WS2812 LED ring, MPU-6050 IMU, VL53L0X time-of-flight sensor, and a TB6612FNG motor driver reading current on micro gear motors — a low-cost sensor suite that gives defect and blockage cues without onboard image processing, keeping the platform's power and compute budget small.